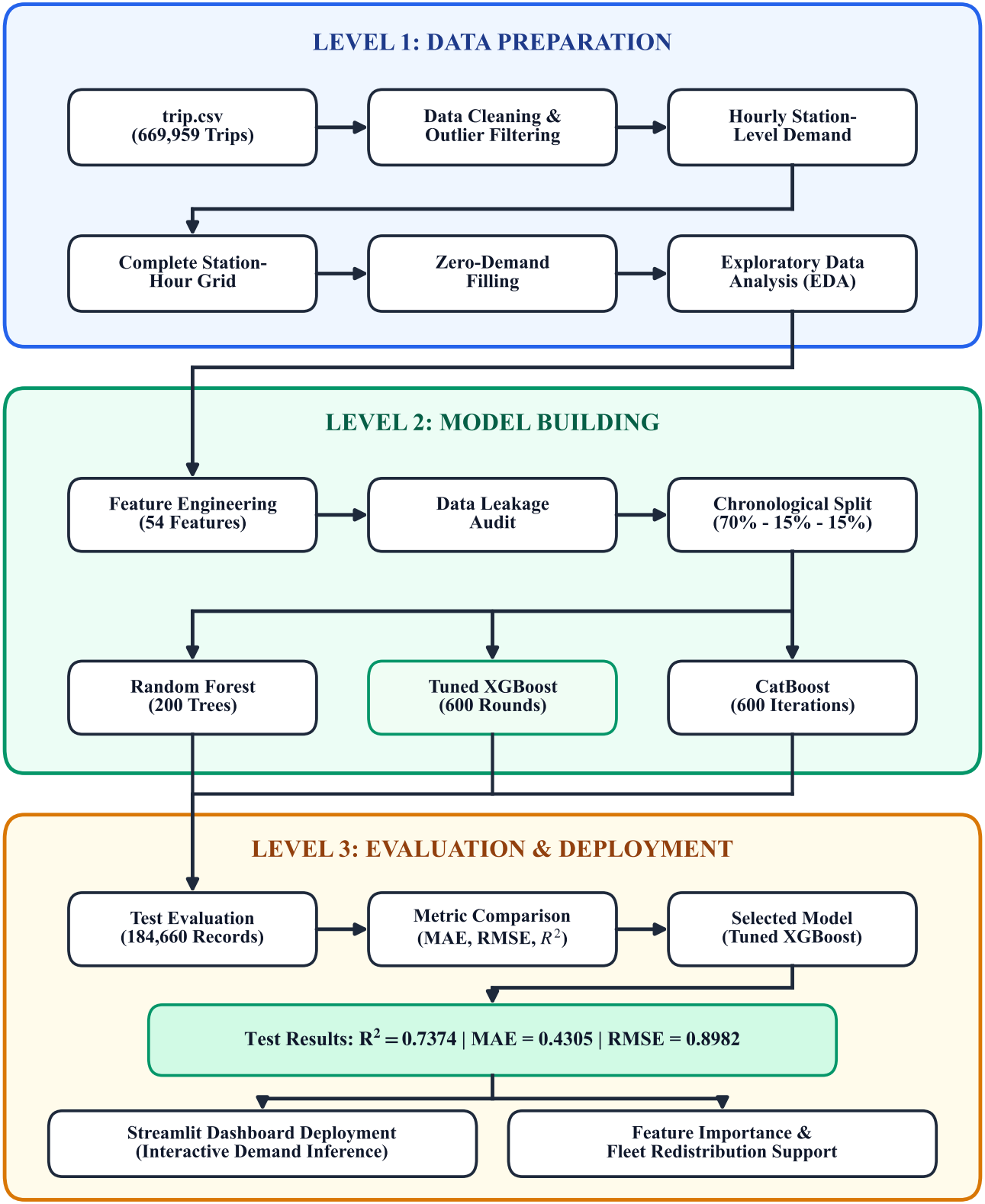
LEVEL 1: DATA PREPARATION
trip.csv (669,959 Trips)
Data Cleaning & Outlier Filtering
Hourly Station-Level Demand
Complete Station-Hour Grid
Zero-Demand Filling
Exploratory Data Analysis (EDA)
LEVEL 2: MODEL BUILDING
Feature Engineering (54 Features)
Data Leakage Audit
Chronological Split (70% - 15% - 15%)
Random Forest (200 Trees)
Tuned XGBoost (600 Rounds)
CatBoost (600 Iterations)
LEVEL 3: EVALUATION & DEPLOYMENT
Test Evaluation (184,660 Records)
Metric Comparison (MAE, RMSE, R^2)
Selected Model (Tuned XGBoost)
Test Results: $R^2 = 0.7374$ | MAE = 0.4305 | RMSE = 0.8982
Streamlit Dashboard Deployment (Interactive Demand Inference)
Feature Importance & Fleet Redistribution Support